

UNIT 6: Mechanical Oscillations and Sound Waves

Subject: Physics

Grade: Grade 9

Core idea: Motion that repeats and travels.

Oscillations

Periodic motion, frequency, period, and an introduction to simple harmonic motion.

Wave Types and Properties

Transverse and longitudinal waves, and key properties: wavelength, frequency, and speed.

Sound Waves

Production and transmission of sound, speed of sound, and echo as reflection of sound waves.

Energy Transfer

How mechanical waves carry energy without transferring matter over long distances.

Mastery Goal

Explain wave behavior and sound using the correct wave concepts.